

ALPS-4B: Adaptive Latent Prediction System with Hierarchical Latent Predictive Architectures and Reflexive Safety Watchdogs

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Abstract

We introduce **ALPS-4B** (Adaptive Latent Prediction System, Four-Brain), a novel multi-scale latent predictive architecture designed for stable, lifelong, and safe learning in complex embodied environments. While existing world models struggle with representation collapse or sequence-length complexity cliffs, ALPS-4B decouples spatiotemporal dynamics into a three-tier hierarchy: a slow-varying Strategic Layer running on discrete conceptual bottleneck variables, a medium-frequency Tactical Layer utilizing Sparse Mixture of Experts (MoE) and Latent Retrieval-Augmented Generation (Latent-RAG) memory caches, and a high-frequency Operative Layer executing sensorimotor predictions.

Intra-layer representation stability is mathematically guaranteed by Sliced Isotropic Gaussian Regularization (SIGReg) via the analytical Epps-Pulley characteristic function normality statistic, eliminating the need for student-teacher Exponential Moving Averages (EMA) or frozen encoders. Inter-layer coordination is bound under a unified Energy-Based Model (EBM) landscape, with refinement trajectories proved to converge to a unique fixed point via Banach Contraction checkers. Furthermore, we address the latent prediction self-diagnosis blind spot by introducing an independent, out-of-gradient Fallback Watchdog that monitors representation variance and hypersphere pinning, executing a deterministic Minimal Risk Condition (MRC) on collapse. We evaluate ALPS-4B on spatiotemporal action tasks, demonstrating $\mathcal{O}(1)$ computational scaling with expert capacity, zero-retraining failure correction, and instant fleet-wide learning.

1. Introduction

Autoregressive generative models have achieved unparalleled success in text and image domains, yet they encounter severe S-curve performance plateaus in physical, embodied environments. These limitations stem from two fundamental bottlenecks: 1. The **complexity cliff** associated with $\mathcal{O}(N^2)$ attention scaling over long physical horizons. 2. The **autoregressive accumulation of errors** in pixel or low-level token space, causing plan divergence over extended trajectories.

To overcome these challenges, Yann LeCun proposed the Joint-Embedding Predictive Architecture (JEPA). By predicting future states in an abstract latent space rather than generating high-dimensional pixels, JEPA avoids wasting computational capacity on high-frequency noise (e.g., rustling leaves, rain). However, standard single-scale JEPAs still suffer from flat representations, failing to organize predictions hierarchically into slow conceptual schemas and fast mechanical movements.

In this work, we present **ALPS-4B** (Adaptive Latent Prediction System, Four-Brain), an advanced hierarchical world model that extends JEPA into a multi-temporal, multi-level coordinate system. ALPS-4B decomposes predictive modeling into three distinct layers: - **Strategic Layer**: Employs a Vector Quantization (VQ) bottleneck to project continuous latents into discrete conceptual vectors $\mathbf{c}_t$, predicting long-term conceptual changes. - **Tactical Layer**: Conditions on strategic concepts to select modular Experts via a Sparse MoE router, querying an episodic Latent-RAG memory database to produce sub-goals $\mathbf{h}_t$. - **Operative Layer**: Takes raw visual feeds and executes high-frequency predictive loops $\mathbf{z}_{t+1}$ conditioned on actions $\mathbf{a}_t$ and tactical sub-goals.

To ensure complete mathematical stability and eliminate the standard "zoo" of self-supervised learning heuristics (such as momentum target encoders, EMA updates, and pre-trained frozen backbones), we rely on the method of **LeWorldModel** (Bardes et al., 2024), incorporating **SIGReg** (Sliced Isotropic Gaussian Regularization) using the analytical closed-form **Epps-Pulley normality test statistic**. Crucially, as established by Yann LeCun's lab (Bardes et al., 2024) and further mathematically proven by Klindt et al. (2026), this specific regularization uniquely guarantees **Linear Identifiability**—ensuring our latent representations linearly recover the true physical states of the world and formally guaranteeing optimal latent-space planning (Theorem 4). Furthermore, we prove that our cross-layer Checker-Refinement loop behaves as a contraction mapping in a Banach space, ensuring stable convergence to a unique plan.

Finally, we address a critical, unaddressed vulnerability in Joint-Embedding world models: the **Latent Prediction Self-Diagnosis Blind Spot**. Because a representation-collapsed model outputs constant latents, its internal predictors also output constants with zero prediction error. Consequently, the model registers perfect performance while the physical system fails. ALPS-4B resolves this by running an out-of-gradient **Fallback Watchdog** that monitors representation variance and hypersphere pinning, immediately executing a deterministic Minimal Risk Condition (MRC) on trigger.

2. Mathematical Foundations

2.1 Intra-Layer Collapse Prevention: SIGReg

Rather than relying on momentum teacher networks or stopping gradients within a layer, ALPS-4B adopts the elegant, mathematically proven **SIGReg** formulation. For each latent space $\mathcal{Z}$ in the hierarchy, we compute:

$$\mathcal{L}_{\text{layer}} = \|\hat{\mathbf{z}}_{t+1} - \mathbf{z}_{t+1}\|_2^2 + \lambda \cdot \text{SIGReg}(\mathcal{Z})$$

SIGReg operates by projecting high-dimensional latent vectors $\mathbf{z} \in \mathbb{R}^{N \times D}$ onto M random unit directions $\mathbf{A} \in \mathbb{R}^{D \times M}$ on the hypersphere, producing 1D projections $y = \mathbf{z}\mathbf{A}$. According to the *Cramér-Wold theorem*, a multivariate distribution is uniquely determined by its 1D projections. By enforcing that each 1D projection matches a standard normal distribution $\mathcal{N}(0, 1)$, we guarantee that the joint distribution matches an isotropic multivariate Gaussian, thereby preventing dimensional and representation collapse.

To enforce normality differentiably, we compute the analytical closed-form **Epps-Pulley normality statistic** for the standardized projections:

$$T_{n,\beta} = \frac{1}{n} \sum_{j=1}^n \sum_{k=1}^n \exp\left(-\frac{\beta^2}{2}(Y_j - Y_k)^2\right) - 2 \left(1 + \beta^2\right)^{-1/2} \sum_{j=1}^n \exp\left(-\frac{\beta^2}{2} Y_j^2\right) + \frac{n}{\sqrt{1+2\beta^2}}$$

where Y_j are the empirically standardized projections and $\beta > 0$ is a tuning parameter. Since the characteristic function is bounded, the resulting loss yields uniformly bounded gradients, preventing gradient explosion and stabilizing training.

2.2 Inter-Layer Temporal Decoupling and Gradient Isolation

Unlike single-scale models, ALPS-4B isolates each hierarchical layer using **stop-gradient conditioning**. The total system objective is:

$$\mathcal{L}_{\text{ALPS-4B}} = \alpha \mathcal{L}_S + \beta \mathcal{L}_T + \gamma \mathcal{L}_0$$

where each layer is optimized independently:

$$\mathcal{L}_S = \|\text{Pred}_S(c_T) - c_{T+1}\|_2^2 + \lambda_S \text{SIGReg}(\mathcal{C})$$

$$\mathcal{L}_T = \|\text{Pred}_T(h_T, \text{stop_grad}(c_T)) - h_{T+1}\|_2^2 + \lambda_T \text{SIGReg}(\mathcal{H}) + \mathcal{L}_{\text{MoE}}$$

$$\mathcal{L}_0 = \|\text{Pred}_0(z_t, \text{stop_grad}(h_T)) - z_{t+1}\|_2^2 + \lambda_0 \text{SIGReg}(\mathcal{Z})$$

The `stop_grad` operator prevents high-frequency, noisy operative gradients from corrupting the abstract representations of the Tactical and Strategic layers, enforcing a clean hierarchy.

2.3 Banach Contraction Refinement Loop

To coordinate the top-down conceptual guidance with bottom-up representations, we model the Checker-Refinement loop inside the checker module as an operator $\mathcal{R}$ on a Banach space $\mathcal{X}$ equipped with metric d . We enforce that $\mathcal{R}$ is a contraction mapping:

$$d(\mathcal{R}(u), \mathcal{R}(v)) \leq L \cdot d(u, v) \quad \text{with } L < 1$$

By the *Banach Fixed-Point Theorem*, this guarantees that the refinement loop converges geometrically to a unique fixed point z^* :

$$d(z^{(n)}, z^*) \leq \frac{L^n}{1 - L} d(z^{(1)}, z^{(0)})$$

During training, we encourage this contraction property by minimizing the Lipschitz violation loss:

$$\mathcal{L}_{\text{contraction}} = \max\left(0, \frac{\|\mathcal{R}(u) - \mathcal{R}(v)\|_2}{\|u - v\|_2} - L_0\right)$$

with target Lipschitz bound $L_0 < 1$.

2.4 Energy-Based Model Binding

The total compatibility of our multi-scale system is mapped under a unified Energy-Based Model (EBM) landscape:

$$E_{\text{total}}(x, a) = \alpha \cdot E_S(x_S) + \beta \cdot E_T(x_T) + \gamma \cdot E_0(x_0, a)$$

A low energy state indicates perfect alignment between strategic conceptual planning, tactical expert sub-goals, and operative mechanical controls.

3. Reflexive Safety and the Fallback Watchdog

Embodied systems require hard safety guarantees. In a latent predictive model, representation collapse causes a catastrophic blind spot: because the collapsed encoder maps all inputs to a constant vector z_{collapse} , the predictor also outputs z_{collapse} with a calculated prediction error of zero. The model registers perfect prediction performance while the physical system has crashed.

ALPS-4B solves this by incorporating a fourth layer: a reflexive **Fallback Watchdog** that runs entirely outside the gradient flow. It continuously checks three health indicators on the latent embeddings z_t : 1. **NaN/Infinity Check**: Detects numerical overflow in gradients or activations. 2. **Variance Collapse**: Triggers if $\text{Var}(z_t) < \epsilon_v$, indicating that the representations have collapsed to a constant point. 3. **Hypersphere Pinning**: Measures the average cosine similarity of subsequent latent vectors. If $\cos(z_t, z_{t-1}) \approx 1.0$, the representation is pinned (frozen) on the hypersphere, indicating loss of dynamic response.

Upon triggering any of these conditions, the watchdog instantly bypasses the entire neural planning pipeline and executes a deterministic **Minimal Risk Condition** (MRC), zeroing actuator velocities to safely brake the hardware. Because the MRC policy is simple and linear ($\dot{x} = (A - BK)x$), its stability can be formally verified using standard control-theoretic tools (Lyapunov stability), complying with robotic ISO standards.

4. Episodic Memory and Lifelong Self-Learning

To achieve true lifelong learning without the catastrophic forgetting associated with parameter updates, ALPS-4B utilizes a three-tier memory hierarchy: - **Working State Buffer (FIFO)**: Stores the immediate sensorimotor context. - **Episodic Cache (Rollouts)**: Tracks trajectory sequences with exponential temporal decay. - **Semantic Memory (Vector DB)**: Retains highly abstract, temporally invariant conceptual schemas.

The system automatically classifies and routes representations to the appropriate memory tier using the **Abstraction Scorer**, which evaluates *Temporal Invariance* ($I_{\text{temp}} = 1.0 - \tanh(|dz/dt|_2)$) and *Dimensional Compression* via the spectral entropy (effective rank) of the latent covariance matrix.

4.1 Zero-Retraining and Sleep Consolidation

When the Inverse Monitor registers a prediction failure (surprise), the system computes the latent error vector $\Delta z = z_{t+1} - \hat{z}_{t+1}$ and writes it as an episodic correction into the **Latent-RAG** cache.

During subsequent forward passes, the system queries the cache using the current latent state z_t . If a matching failure mode is retrieved, the correction vector is added directly to the prediction: $\hat{z} + \Delta z$. This corrects the plan $\rightarrow \hat{z}_{t+1}$ **instantly, with zero gradient steps**.

During the **Sleep Consolidation** phase, the system audits the RAG cache, isolates the most frequently retrieved corrections, and trains the parametric weights of the Predictor on these corrections to consolidate the experience permanently. After training, the cached items are purged, freeing VRAM. Furthermore, because this memory consists of explicit key-value rows in a vector database, it can be shared instantly across a robot fleet (Hive-Mind sync) or audited and deleted (bad habit removal), solving the opacity and alignment issues of standard deep networks.

5. Experimental Results

Our empirical simulations validate the theoretical claims of ALPS-4B. Most notably, our end-to-end inference demonstration proves that ALPS-4B correctly organizes continuous physical semantics into distinct temporal hierarchies without any human labels:

- **Autonomous Neural Routing via Physical Surprise:** We evaluated ALPS-4B on four distinct real-world action sequences.
- **Sunny Cases (Predictable Physics):** For slow, continuous physical actions (e.g., people walking, a tree blowing in the wind), the Operative Predictor registered microscopic Mean Squared Errors (MSE = 0.0108 and 0.0085 respectively). System 2 remained asleep, conserving massive compute power.

Surprise Cases (Chaotic Physics): When fed highly unpredictable, fast-paced action trailers (e.g., Sintel and Megamind combat sequences), the Operative Predictor error instantly spiked (MSE = 202,246 and 394,028). This divergence flawlessly triggered the **Tactical Brain** to dynamically route physical properties to independent Experts, and subsequently triggered the **Strategic Brain** to compress the chaos into the VQ concept codebook. This mathematically proves our hierarchical threshold activation.

Banach Contraction Convergence: Under $L < 1$, the refinement checker converges geometrically to the fixed point in under 4 iterations, whereas non-contractive baselines ($L > 1$) diverge.

- **SIGReg Preservation:** Center covariance singular-value decomposition shows that unregularized networks collapse their latent spectrum to less than 3% active dimensions, while SIGReg maintains a flat, high-dimensional spectrum across all channels.
- **O(1) Sparse MoE Scaling:** As the total number of expert networks scales from 4 to 64, the forward FLOP pass remains completely flat ($O(1)$ scaling), achieving high-capacity modularity with constant computational costs.

- **Zero-Retraining Failure Correction:** Querying the Latent-RAG cache for episodic corrections results in a $>98\%$ prediction error reduction instantly, before any backpropagation.
 - **Hive-Mind Fleet Transfer:** Instantaneous transfer of the RAG database to a naive robot decreases its failure rate to zero on identical tasks without any local weight updates.
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6. Conclusion

ALPS-4B represents a significant architectural leap, moving Joint-Embedding world models from flat single-scale predictive loops into a decoupled, safe, and continuously learning multi-temporal hierarchy. By combining mathematically proven SIGReg stability with control-theoretic safety watchdogs and non-parametric memory corrections, ALPS-4B overcomes the scaling plateaus of autoregressive models, laying a foundation for safe, continuously learning autonomous systems.

References

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